

A Database-Driven Analysis of Movie Evaluation Across IMDb, Rotten Tomatoes, and the Oscars

Group 1

1. Introduction

1.1 Background

There are multiple movie platforms and evaluation systems (such as IMDb, Rotten Tomatoes, and major award institutions), each with different standards, audiences, and evaluation criteria. Because of these differences, it is often unclear how much a movie rating actually reflects its overall quality, popularity, or critical recognition.

As a result, people may find it difficult to interpret movie ratings in isolation. By combining audience ratings, popularity measures, and award recognition, we aim to better understand what movie ratings represent and how different forms of recognition relate to each other.

This topic matters because movie ratings influence audience decisions, industry reputation, and even financial success, yet their meaning may vary across platforms.

1.2 Research Questions

1. For movies released from 2016 to 2025, is there a relationship between IMDb ratings and Rotten Tomatoes audience scores?
2. Does Oscar recognition significantly predict IMDb audience ratings after controlling for popularity and genre for movies released between 2016–2025?
3. For movies released from 2016 to 2025, is IMDb popularity, measured by number of user votes, associated with the level of critic engagement on Rotten Tomatoes?

1.3 Impact

This project is useful for several groups. For general audiences, it helps explain whether ratings from different platforms tell a similar story about movie quality. For filmmakers and studios, it provides insight into how audience ratings, popularity, and award recognition are related. For researchers, it offers a clearer way to compare crowd-based ratings, critic evaluations, and institutional recognition within the movie industry.

2. Data

2.1 Datasets

1. IMDB Official Dataset: <https://datasets.imdbws.com/>

This dataset is chosen because it provides standardized audience ratings (0–10 scale) and popularity measure (number of votes). By using that, it directly supports our three research questions. Also, IMDb is a globally recognized and authoritative movie database. The dataset is officially published for research use, increasing reliability.

2. Rotten Tomatoes Critics and Audience Scores Dataset (Kaggle): <https://www.kaggle.com/datasets/matepapava/rotten-tomatoes-critics-and-audience-scores>

This dataset is chosen because it provides evaluations from critics and audiences, and the numerical scores are suitable for relevance and comparison. Since IMDb’s ratings range from 0 to 10, Rotten Tomatoes’ audience ratings (from 0 to 100) will be classified as 0 to 10. After that, it allows us to directly compare it with the IMDb score. It directly supports our three research questions. Although the data is from Kaggle, the original is from the api of Rotten Tomatoes, so it is reliable. It reflects the publicly available comprehensive score and is widely used in academic projects.

3. The Oscar Award Dataset: <https://www.kaggle.com/datasets/unanimad/the-oscar-award>

This dataset is chosen because it enables us to make comparisons between films that have received Oscar recognition and those that haven’t. This dataset directly supports research question 2. Although it is from Kaggle, it is a publicly accessible historical record crawled from the official database of the Oscars.

By combining these datasets, the project can analyze the relationship between audience ratings, popularity, critic evaluation, and award recognition, which cannot be achieved using a single dataset.

2.2 Data Cleaning Steps

For the IMDb Dataset:

Missing values represented as “\N” were converted to proper null values. The columns startYear, averageRating, and numVotes were converted to numeric types. The dataset was filtered to include only movies released between 2016 and 2025. Rows with missing values in key variables (primaryTitle, startYear, averageRating, numVotes) were removed. To ensure reliability, only movies with at least 2,000 votes were retained. Duplicate records were

removed using the movie identifier (tconst), and movie titles were standardized by converting to lowercase and removing punctuation. The cleaned dataset was saved as `movies_filters.csv`.

For the Rotten Tomatoes Dataset:

The dataset was loaded from JSON and converted into a pandas DataFrame. Only relevant columns (title, year, audience_score, critics_score, type) were retained, and rows with missing values were removed. The dataset was filtered to include movies released between 2016 and 2025. Scores were rescaled from 0–100 to 0–10 to align with IMDb ratings. Only entries with type = “movie” were kept. Movie titles were standardized by converting to lowercase and removing subtitles after colons. The cleaned dataset was saved as `clean_rotten_tomatoes.csv`.

For the Oscar Award Dataset:

We retained only the columns relevant to our research questions, including Category, Year, Film, and FilmId. Rows with missing FilmId were removed since they cannot be linked to the IMDb dataset. The dataset was then filtered to include films released between 2016 and 2025. Duplicate records were removed using FilmId to ensure each movie appears only once. After cleaning, the dataset was reduced from 12,014 to 494 rows and saved as `clean_oscar.csv`.

2.3 Exploratory Data Analysis

In the IMDb ratings and voting numbers, we observed some outliers, especially for those movies that were extremely popular or had abnormally low ratings. By deleting incomplete records to handle lost data, this improves data consistency, but it may reduce the overall sample size. Overall, the dataset reveals clear patterns. The IMDb score is positively correlated with the Rotten Tomatoes score. Films that are recognized by the Oscars tend to have higher scores.

2.4 Important Limitations or Biases in the Data

- Selection bias: Filtering movies with at least 2,000 votes excludes less popular films
- Matching issues: Title-based matching across datasets may introduce errors
- Different scoring systems: IMDb (0–10) vs Rotten Tomatoes (originally 0–100)
- Missing data removal: Dropping missing values reduces dataset size
- Oscar bias: Oscar-recognized films may already represent higher-quality productions

3. Methodology

3.1 Analytical Framework

We used Python to execute queries in both Oracle and MongoDB and retrieve the relevant data. The queries filtered movies from 2016 to 2025 and extracted variables needed for each analysis. The results were then processed using pandas and visualized with matplotlib, allowing us to combine database querying and data analysis in a clear and reproducible workflow.

3.2 Variables and Research Design

Our analysis examines the relationships among audience ratings, popularity, critic evaluations, and award recognition. Each research question was translated into measurable variables from the datasets.

For Research Question 1, we compared IMDb average ratings and Rotten Tomatoes audience scores to evaluate their consistency. For Research Question 2, we examined the relationship between Oscar recognition and IMDb ratings while considering popularity and genre. For Research Question 3, we analyzed whether IMDb popularity, measured by the number of user votes, is associated with Rotten Tomatoes critic scores.

3.3 Analytical Methods and Visualization

We applied descriptive and exploratory methods, including filtering, joins, and aggregation, to identify patterns and relationships across movies. The analysis focuses on associations rather than causal inference.

Visualizations were used to support interpretation. For Research Question 1, boxplots compare rating distributions, while hexbin plots show density and overall trends between platforms. For Research Question 2, scatter plots examine the relationship between ratings and popularity, and grouped line charts based on binned popularity and genre enable comparison between Oscar and non-Oscar movies. For Research Question 3, scatter plots with a fitted trend line (on a log scale) illustrate the relationship between popularity and critic scores, while binned line charts summarize average trends across popularity ranges.

These techniques highlight both distributional differences and relationships in a clear and interpretable way.

4. Database Implementation

4.1 Oracle Schema Design

We designed a relational schema in Oracle with three tables: *IMDb_Movies*, *Oscar_Awards*, and *Rotten_Tomatoes*. The *IMDb_Movies* table serves as the central entity, using *tconst* as the primary key and storing core attributes such as *primaryTitle*, *startYear*, *averageRating*, *numVotes*, and *genres*.

The *Oscar_Awards* table records award information, including *FilmId*, *Year*, and *Film*, and is directly linked to *IMDb_Movies* through the shared identifier *tconst*. The *Rotten_Tomatoes* table stores audience and critic scores (*audience_score*, *critics_score*) along with *title* and *year*, which together form a composite key and are used to match records with *IMDb_Movies*.

4.2 MongoDB Document Design

In MongoDB, we redesigned the data using a document-oriented model. Each movie is represented as a single document that contains all relevant information, including IMDb attributes, Rotten Tomatoes scores, and Oscars data.

The document structure groups related data together. For example, rating information from different sources is stored as nested fields within the same document. This creates a self-contained representation of each movie, where all attributes can be accessed without combining multiple collections.

A key design decision was to use embedded documents instead of document references. Since the data naturally centers around a single entity (a movie), embedding allows related attributes to be stored together in a hierarchical structure. This reduces the need to manage relationships across multiple collections.

The resulting document model reflects the real-world structure of the data, where each movie acts as a unified object containing multiple types of information.

4.3 Initial Design Evaluation

Before implementation, MongoDB appeared to be the more natural fit, as each movie combines information from multiple sources. Representing each movie as a single document aligns with the structure of the data and provides a more intuitive model than separating information across multiple tables.

5. Database Comparison and Reflection

5.1 Final Decision

After completing the project using both Oracle and MongoDB, we ultimately chose Oracle as the database for this project.

This is because the project integrates data from three different sources: IMDb, Rotten Tomatoes, and the Oscars dataset. Much of our work involves matching records across data sources, filtering movies by year, grouping data, and comparing relationships between different variables. These tasks are more straightforward in Oracle because its relational structure makes it easier to write and understand join and group queries.

MongoDB also has its advantages for this project, especially since our data is organized around the “movie” as the central entity. By storing each movie as a document and embedding rating and award-related information within it, MongoDB provides a more natural and compact data structure. For simple queries that only need to retrieve fields related to a single movie, MongoDB is more convenient because it does not require join operations. However, when queries become more analytical and require clear comparisons across different data sources, Oracle is easier to manage.

If the project were to include more reviews, text content, or a large number of optional fields for each movie, MongoDB would be the better choice. In that scenario, the document-based model would better suit the data. However, since our current project emphasizes structured comparisons across multiple datasets, Oracle is more appropriate.

5.2 Applicability of Each Research Question

For each research question, different databases have their respective advantages. Research Question 1 is slightly easier to address in MongoDB, as IMDb ratings and Rotten Tomatoes audience scores can be stored directly within the same document. Research Question 2 is easier to handle in Oracle, as it requires more structured comparisons centered on Oscar recognition, popularity, and genre. Research Question 3 can be addressed by both systems, but Oracle offers clearer organization of comparison logic, while MongoDB is more convenient for directly extracting fields.

5.3 Comparison of Query Difficulty

Overall, MongoDB is better suited for simple, single-movie queries, while Oracle is better suited for more structured, analytical queries. Since our project is inherently more focused on analysis and comparison, Oracle will be our final choice.

6. Results

6.1 Research Question 1

How do the distributions of IMDb ratings and Rotten Tomatoes audience scores compare for movies released between 2016 and 2025?

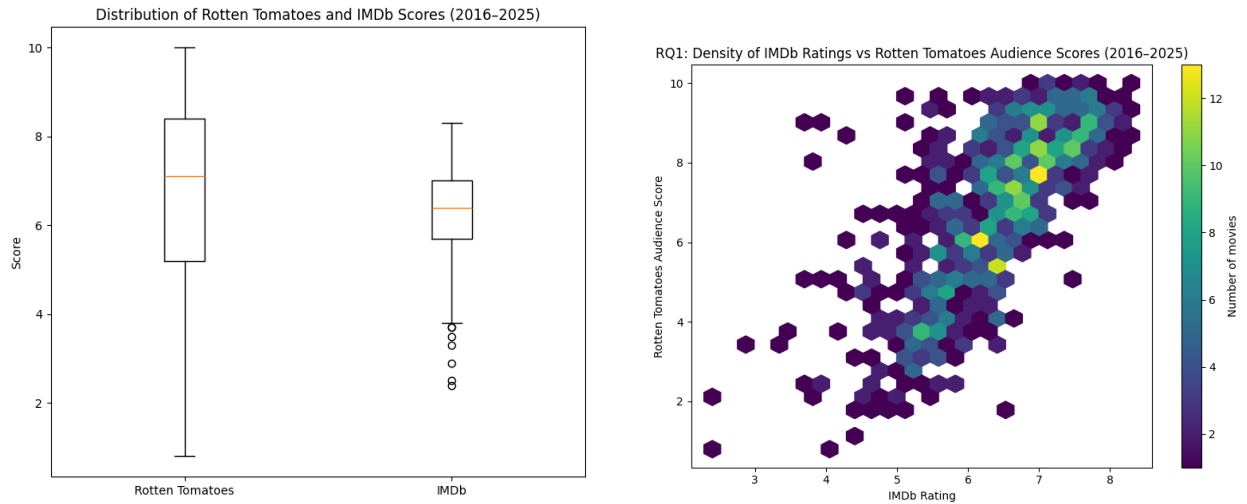


Figure 1: Research Question 1 visualization

Explanation and Discussion

The boxplot shows that Rotten Tomatoes scores have a wider distribution than IMDb ratings, with a slightly higher median and more extreme values. In contrast, IMDb ratings are more concentrated, mostly between 5.5 and 7.

The hexbin plot reveals a clear positive relationship between the two platforms, with most movies clustered around IMDb scores of 6 to 7.5 and Rotten Tomatoes scores of 6 to 8. However, there is noticeable variation, as movies with similar IMDb ratings can have different Rotten Tomatoes scores.

The boxplot alone cannot show agreement between the two platforms, but the hexbin plot suggests that they are generally aligned. Overall, there is a positive relationship between IMDb ratings and Rotten Tomatoes audience scores, but the relationship is not perfect. However, the spread in the data shows that some movies receive different evaluations on the two platforms.

6.2 Research Question 2

Does Oscar recognition significantly predict IMDb audience ratings after controlling for popularity and genre for movies released between 2016–2025?

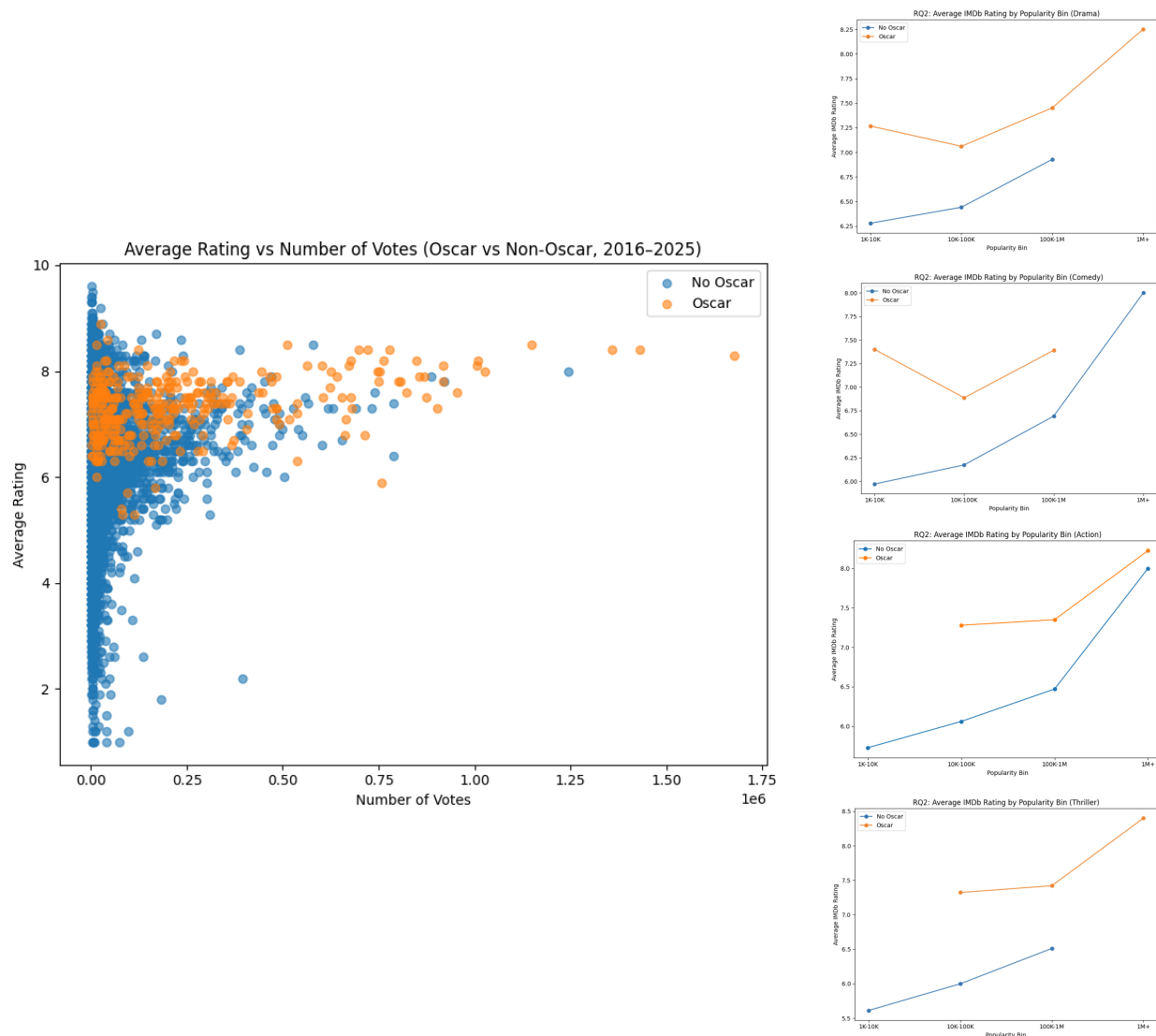


Figure 2: Research Question 2 visualization

Explanation and Discussion

The scatter plot shows a weak positive relationship between IMDb ratings and popularity, with a large spread in ratings, especially among less popular movies. Oscar-recognized films tend to appear more frequently in the higher rating range.

The grouped line charts provide clearer evidence after controlling for popularity and genre. Across all genres (comedy, drama, thriller, and action), Oscar-recognized movies consistently

have higher average IMDb ratings than non-Oscar movies within the same popularity bins. Overall, these results strongly suggest that Oscar recognition is positively associated with higher IMDb ratings even after controlling for popularity and genre.

6.3 Research Question 3

For movies released from 2016 to 2025, is IMDb popularity (number of user votes) associated with the level of critic engagement on Rotten Tomatoes?

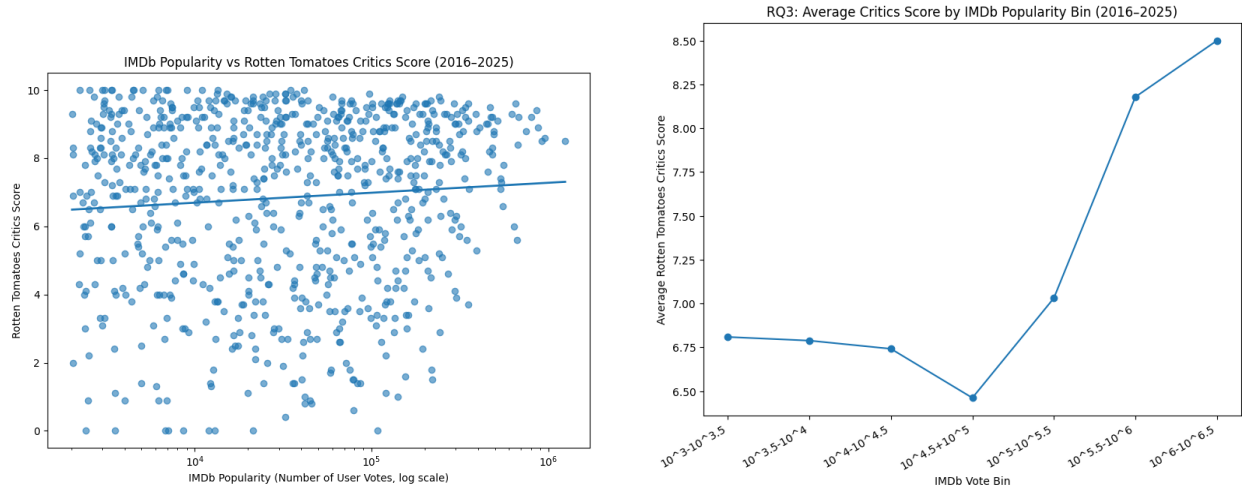


Figure 3: Research Question 3 visualization

Explanation and Discussion

The scatter plot shows the relationship between IMDb popularity (number of votes, on a log scale) and Rotten Tomatoes critics scores. There is a slight upward trend, as indicated by the fitted line, suggesting that more popular movies tend to have somewhat higher critic scores. However, the data points are widely spread across all levels of popularity. Movies with similar numbers of votes can have very different critic scores, and high critic scores also appear among less popular movies.

The binned line chart groups movies into popularity ranges. For lower popularity bins, the average critics score stays relatively stable or even slightly decreases. As popularity increases, the average critics score increases more noticeably, especially in the highest popularity ranges.

Overall, there is a weak positive association between popularity and critic scores. While highly popular movies tend to have somewhat higher scores, popularity is not a strong or consistent predictor of critic evaluation.

7. Limitations and Future Work

7.1 Limitations

This study has several limitations. First, the project integrates data from IMDb, Rotten Tomatoes, and the Academy Awards datasets; since movie titles and formats are not always fully consistent, there may be mismatches between different data sources. Second, we removed missing values and filtered out movies with fewer than 2,000 IMDb votes; while this improved data quality, it may also introduce selection bias due to the underrepresentation of less popular films. Third, ideally, both the Oscars and Rotten Tomatoes should be subject to a total participation constraint; however, in practice, both Rotten Tomatoes and the Oscars include films not found in the IMDb database. Finally, the study primarily employs descriptive analysis, so while we can identify correlations, we cannot draw causal inferences.

7.2 Future Work

If we had more time and resources, we would refine the cross-dataset matching process and incorporate additional variables, such as film budgets, box office performance, and distribution platforms. We would also apply more rigorous statistical models to test whether the observed correlations hold true after controlling for other factors. Furthermore, future research could explore whether these correlations differ across film genres, or whether the link between critics' acclaim and awards is stronger than that based on audience ratings.

References

IMDb. (2024). *IMDb non-commercial datasets*. Retrieved from <https://datasets.imdbws.com/>

Papava, M. (2024). *Rotten Tomatoes: Critics & audience scores*. Kaggle. Retrieved from <https://www.kaggle.com/datasets/matepapava/rotten-tomatoes-critics-and-audience-scores>

Fontes, R. (2025). *The Oscar award, 1927–2025*. Kaggle. Retrieved from <https://www.kaggle.com/datasets/unanimad/the-oscar-award>

AI Tool Use Declaration

We did not use AI for assistance.